

IAL Biology Unit 2 (WBI12): Rapid Revision

Cells, Development, Biodiversity and Conservation · last-minute summary · 1 h 30 min · 80 marks · usually 8 questions · more recall-heavy than Unit 1: “give / state / name” is worth about 9 marks a paper

HOW THE MARKS WORK

- **Guaranteed in all 21 papers:** cell ultrastructure, gametes and fertilisation, maths and data skills.
- **Near-certain (19/21):** prokaryotes vs eukaryotes, microscopy, mitosis, plant cell structure.
- **Barely tested:** evolution. Natural selection appeared in only 2 of 21 papers and the word “evolution” in none. **Adaptation and niche** appear in 14-17 — revise those instead.
- The highest-frequency content words in the mark schemes are “**cellulose**” (×76) and “**microfibril**” (×40).
- **Magnification** is the commonest single calculation; **mitotic index** appears in 12 of 21.

CELL ULTRASTRUCTURE

Organelle	Function — as the scheme wants it
Nucleus	Contains DNA; nuclear pores allow mRNA out
Nucleolus	Makes ribosomes / rRNA
Mitochondrion	Aerobic respiration / ATP synthesis; cristae give a large surface area
rER	Ribosomes on the surface; protein synthesis and transport
sER	Lipid and steroid synthesis
Golgi	Modifies proteins, packages them into vesicles ; makes lysosomes
Lysosome	Hydrolytic enzymes ; digests material
Ribosome	80S in eukaryotes, 70S in prokaryotes; protein synthesis
Centriole	Forms the spindle
Chloroplast	Photosynthesis; thylakoids, stroma, 70S ribosomes, DNA loop

THE PROTEIN SECRETION ROUTE

Asked almost every session as a sequence: **nucleus** (gene transcribed; mRNA leaves through a nuclear pore) → **ribosome on the rER** (translated) → **rER** (folded, transported) → **vesicle** → **Golgi** (**modified** and packaged) → **vesicle** → **cell surface membrane** → released by **exocytosis**. Mitochondria supply the **ATP**.

Prokaryote	Eukaryote
No nucleus — circular DNA, free in the cytoplasm	Nucleus with a double membrane; linear DNA with histones
No membrane-bound organelles	Mitochondria, ER, Golgi, lysosomes
70S ribosomes	80S ribosomes
Peptidoglycan (murein) wall	Cellulose wall in plants; none in animals
Plasmids; smaller (0.5-5 µm)	No plasmids; larger (10-100 µm)

Plant vs animal: plants have a **cellulose cell wall**, chloroplasts, a **permanent vacuole with tonoplast**, and store starch; animals have centrioles and store glycogen.

MICROSCOPY

	Light vs electron
Resolution	Light ~200 nm ; TEM ~0.1-0.5 nm . Electrons have a shorter wavelength than light , which is why resolution is higher
Specimen	Light: living or dead, colour, cheap. EM: dead only, in a vacuum , black and white, expensive, complex preparation
TEM vs SEM	TEM: thin section, internal detail , 2D. SEM: surface, 3D , lower resolution

Resolution = the ability to distinguish two points as separate. **Magnification** = how many times larger the image is. **Artefact** = a structure produced during preparation that is not present in the living cell.

Magnification = image ÷ actual. Convert to the same units first (1 mm = 1000 µm = 10⁶ nm). With a graticule, calibrate against a stage micrometer.

THE CELL CYCLE AND MITOSIS

Interphase: G₁ growth, **S phase DNA replication**, G₂ growth. Then mitosis, then cytokinesis.

Stage	What happens
Prophase	Chromosomes condense and become visible; nuclear envelope breaks down ; spindle forms
Metaphase	Chromosomes line up at the equator , attached by centromeres to spindle fibres
Anaphase	Centromeres divide ; sister chromatids pulled to opposite poles by shortening spindle fibres
Telophase	Chromosomes decondense; nuclear envelopes re-form

Mitosis produces two genetically identical diploid daughter cells — growth, repair, asexual reproduction. **Meiosis** produces four genetically different haploid cells; variation comes from **crossing over** in prophase I and **independent assortment** in metaphase I.

Mitotic index = (number of cells in mitosis ÷ total number of cells) × 100. Count a stated large number of cells; a higher index means a greater proportion of cells are dividing.

STEM CELLS AND DIFFERENTIATION

Totipotent: can form **any** cell type **including** extra-embryonic tissue. **Pluripotent**: any cell type of the body but not extra-embryonic. **Multipotent**: a limited range (adult stem cells).

Differentiation: all cells contain the **same genes**; different genes are **switched on or off** → different **mRNA** → different **proteins** → different structure and function. Say “**gene expression**”, not “genes are lost”.

Ethics: give **both sides** — therapeutic potential and the reduction of suffering, against the status of the embryo and the availability of adult or induced pluripotent alternatives.

PLANT CELLS, CELLULOSE AND SUSTAINABILITY

CELLULOSE AND THE PLANT CELL WALL

β-glucose; **alternate molecules are inverted** so the OH groups align; **1,4 glycosidic bonds**; long **straight, unbranched** chains; held side by side by **hydrogen bonds** into **microfibrils**, which are bundled into **fibres** and embedded in a matrix of pectins and hemicelluloses. This gives **high tensile strength**, resists **turgor** pressure and prevents the cell bursting.

Plant fibres (sclerenchyma, phloem fibres): cellulose microfibrils and **lignified** walls give strength and stiffness. **Xylem**: dead, hollow, lignified, no end walls — transport and support.

Tensile strength practical: hang masses from a fibre until it breaks; control fibre length and diameter; repeat and take a mean; identify anomalies.

Sustainability: plant fibres and bioplastics are **renewable**, biodegradable and need less energy to produce than oil-based materials; but they may need land, water and fertiliser that compete with food crops. **Starch-based plastics** are made from a renewable resource and decompose.

Mineral deficiencies: nitrate for amino acids and pro-

teins; magnesium for **chlorophyll** — deficiency causes **chlorosis** / yellow leaves; calcium for **calcium pectate** in the middle lamella.

DRUG TESTING AND ANTIMICROBIALS

The historical chain: Digitalis / William Withering → the modern protocol. Modern testing: **pre-clinical** on cells and animals (toxicity), then **Phase I** healthy volunteers (safety, dose), **Phase II** small numbers of patients (efficacy), **Phase III** large randomised trials against existing treatment.

Placebo = a treatment with no active ingredient, so the effect of expectation is controlled. **Double-blind** = neither patient nor doctor knows the allocation, so neither can bias the result. **Randomised** = assignment is not influenced by the investigator.

Antimicrobial practical: sterile technique; paper discs soaked in equal volumes of each solution; **measure the diameter or area of the clear zone**; control disc with solvent only; incubate at a stated temperature; repeat and take a mean. Zones are **not always circular**, which is why area is better than a single diameter.

BIODIVERSITY, ADAPTATION AND CLASSIFICATION

Term	Definition the scheme accepts
Biodiversity	Number of different species (species richness) and the genetic diversity within a species — two components
Species richness	The number of species present
Niche	The role an organism plays in its habitat — plus a stated reason when a context is given
Endemic	Found only in that one place

Three kinds of adaptation, and the question usually wants one of each: **anatomical** (structure), **physiological** (internal processes), **behavioural** (what it does). Always say **how the feature helps survival**.

Classification: domain, kingdom, phylum, class, order, family, genus, species. **Binomial** = genus + species, italicised or underlined. Molecular evidence — **DNA base sequences, mRNA and amino acid sequences** — gives a more objective comparison than morphology; **the more similar the sequence, the more closely related**.

Species = a group of organisms that can interbreed to produce **fertile** offspring.

Natural selection (rare, but learn the chain): **random mutation** produces a new allele → **variation** → a **selection pressure** acts → individuals with the advantageous allele **survive and reproduce** → they **pass the allele to their offspring** → **allele frequency increases over generations**.

CONSERVATION

Seed banks: store many seeds in a small space; **dried and cooled** to reduce metabolic rate so they remain viable for longer; cheaper than growing plants; preserve genetic diversity. Limitations: not all species produce storable seeds; **viability must be tested periodically** by germinating a sample; no evolution in storage.

Zoos: captive breeding, **studbooks** to avoid inbreeding, reintroduction, research and education. Problems: small populations risk **inbreeding depression** and **loss of genetic diversity**; animals may lack the behaviour needed to survive if released; cost.

Sampling: random quadrats using random coordinates **to avoid bias**; measure density, frequency or percentage cover; **transects** for a gradient; take enough samples (species-area curve). Population = mean per quadrat $\times$ (total area $\div$ quadrat area).

THE REJECTIONS BANK

Do not write	Write instead
"Cells lose genes as they specialise"	All cells have the same genes ; different genes are expressed
"Electron microscopes magnify more" as the whole answer	They have higher resolution because electrons have a shorter wavelength
"Cellulose is strong"	Hydrogen bonds between straight β -glucose chains form microfibrils giving tensile strength
Niche as "where it lives"	The role it plays, with a reason
Biodiversity as "number of species" only	Species richness and genetic diversity within species
"The selection pressure causes the mutation"	Mutation is random and comes first
"Immune" to a chemical	Resistant
Magnification without converting units	Convert both measurements to the same unit first
"Seeds last forever in a seed bank"	Viability must be tested periodically

WALKING INTO THE EXAM

This paper rewards precise recall, so give the exact term. Name the organelle, the bond, the stage. On "describe" questions do not explain, and on "explain" questions do not merely describe. Convert units before every magnification calculation, and quote data with units.